

Sign Language Recognition using Temporal Classification

Course: UE24CS352A Machine Learning Mini-Project (Problem 118) | Team 83 | Ref: Cate, Dalvi, Hussain (2015)

1. Problem Statement & Motivation

Camera-free sign language recognition through wearable sensor gloves provides invariance against lighting, occlusion, and background clutter. This project reproduces and extends the 95-sign Auslan (Australian Sign Language) recognition benchmark from Cate, Dalvi, and Hussain (2015). The original study suffered poor performance on low-quality data and reported a total failure of standard LSTMs (F1 = 0.066). We aim to: (1) reproduce and benchmark classical baselines, (2) perform rigorous feature ablation to establish modality hierarchy, and (3) diagnose and resolve the structural flaw in the paper's LSTM.

2. Dataset & Cleaning Pipeline

We utilize Mohammed Waleed Kadous's Auslan benchmark (UCI ML Repository #114). While the UCI repository lists a 200 Hz high-quality dual-glove dataset, its archive links return broken symlinks and 403 Forbidden HTTP errors; hence all work is rigorously evaluated on the low-quality Nintendo PowerGlove dataset (50 Hz, 1 hand, across multiple signers). **Cleaning applied:** (1) Removed 3 non-sign calibration files ('cal-*) and 1 corrupted empty file. (2) Stripped hex status bytes. (3) Out of 11 numeric channels, cols 4 and 5 are zero-variance constants and col 10 is an exact duplicate of col 9. Pruning leaves **8 active features**: POS (x, y, z; cols 0-2), ROT (roll; col 3), and finger bends F1-F4 (cols 6-9). The cleaned corpus contains **6,648 recordings across 95 classes** (~70 per class, mean duration 58.4 frames).

3. Methodology & Feature Engineering

Temporal Normalization: Variable duration signs are standardized to T = 57 frames (dataset mean) via Fourier resampling (scipy.signal.resample). **Spatial Scaling:** We evaluate per-example min-max scaling ([0, 1] per channel per sign) versus global dataset scaling. **Representations:** Static flattened vectors (57x8 = 456 dimensions) for SVM/Logistic Regression, and sequential tensors (N, 57, 8) for recurrent networks.

Fixing the Paper's LSTM: The paper formulated sign recognition using Mean Squared Error (MSE) evaluated at every discrete frame, which forces intermediate frame predictions where gestures are non-discriminative. We resolve this by enforcing sequence-to-label aggregation (return_sequences=False, backpropagating strictly from the final hidden state) paired with 95-way Softmax Sparse Categorical Cross-Entropy, dropout (p=0.3), and validation early stopping.

4. Experimental Results & Analysis

All models are benchmarked under a stratified 70/30 train/test split (seed 42; 4,653 train / 1,995 test instances).

Model	Scaling	Test Acc	Macro F1	Paper F1
Linear SVM (C=1.0)	Per-example	0.306	0.308	—
Linear SVM (C=0.1)	Per-example	0.369	0.365	0.549
Logistic Regression	Per-example	0.385	0.384	0.436
RBF SVM (C=10)	Global	0.576	0.575	—
RBF SVM (C=10)	Per-example	0.602	0.601	0.549
Baseline LSTM (128)	Per-example	0.383	0.370	0.066
Bidirectional LSTM (128)	Per-example	0.437	0.431	—
Stacked LSTM (2x128)	Per-example	0.447	0.440	0.066

4.1 Feature Modality Ablation (RBF SVM)

Condition	Channels Retained	Test Error	Significance
Full Feature Set	POS + ROT + F1-F4 (8 cols)	39.8%	Baseline (Acc: 60.2%, F1: 0.601)
Without POS	ROT + F1-F4 (5 cols)	75.6%	+35.8% error surge: primary trajectory signal
Without ROT	POS + F1-F4 (7 cols)	41.8%	+2.0% error: minor wrist roll contribution
Cumulative Drop	Removed up to F3 (only F4 left)	95.6%	Near-chance baseline without spatial channels

4.2 Error Analysis & Modality Insights

Inspection of off-diagonal confusion matrix elements reveals top confusions: *man* ↔ *please* (5 errors), *which* ↔ *maybe* (5 errors), and *surprise* ↔ *more* (5 errors). These pairs trace nearly identical gross hand paths, differing solely in subtle thumb flexion or finger curvature that is susceptible to PowerGlove sensor quantization.

5. Conclusions & Discussion

- **Literature Reproduction:** Achieved 60.1% Macro F1 with RBF SVM, surpassing the 2015 baseline of 54.9%.
- **LSTM Rectification:** Demonstrated that replacing frame-wise MSE with final-step softmax cross-entropy resolves the paper's reported failure (surpassing 0.066 by an order of magnitude).
- **Model Selection Trade-off:** On small sample sizes (~49 train instances/class across 95 classes), non-linear kernel SVM on Fourier-flattened representations is remarkably competitive with deep recurrent models, while requiring drastically lower training compute.
- **Sensor Hierarchy:** 3D spatial position is the primary classifier driver; finger flexion resolves local gesture ambiguities.